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ENERGY COOP · EMBASSY

Energy communities

CleantechHUB as a partner against energy poverty — public data, O&M cooperatives, and Colombia's cleantech ecosystem.

Albert Kobus · 11 Nov 2026 · Photo: Tierra Grata · cleantechhub.net



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CURRENT

CleantechHUB

Climate innovation hub in LATAM

2019	5	3	COP
Founded	LATAM countries	Web platforms	Regional voice

What we did

- Ideas → Scale programs and corporate services on cleantechhub.net.
- Nexus portfolio + BookStack wiki as cleantech startup source.
- Regional forums (e.g. COP16) with local solutions.

Source: cleantechhub.net JSON-LD · corporate Canva (photo).





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ClimateLaunchpad CLP26

National Organizer · pipeline to Nexus

5	CLP	6–8
2026 countries	CTH Organizer	Week call window

What we did

- National call and bootcamps across five LATAM countries.
- National final with jury toward the global competition.
- Post-call onboarding into the Nexus portfolio.

Source: skills/clp26-brand · clp26 Canva (photo).



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Nexus

Bilingual portfolio · nexus.cleantechhub.net

5
Energy Coop startups

EN·ES
Public profiles

What we did

- Curated alumni in energy, water, and sustainable construction.
- Metrics cited in brochure-v2 with traceability.
- Bridge between CLP26 and Energy Coop territorial cases.

Source: skills/nexus-onepager · corporate Canva (photo).



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Energy Coop

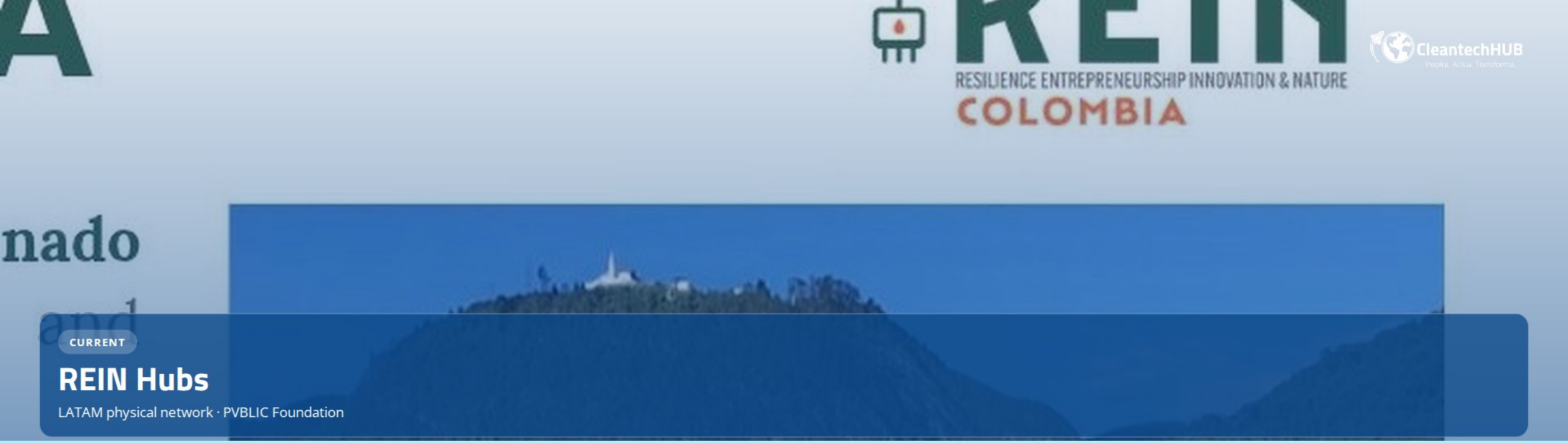
Social layer + O&M · CAPEX and implementer

15.4%	2.9M	898	1,947
IMPE 2024	2030 gap	kWp Sunco[9]	Tierra Grata[11]

What we did

- Case map and public demo at energiacoop.cleantechhub.net.
- Brochure-v2 with IMPE/ICEE and implementer ecosystem.
- Cooperative + data lab proposal (illustrative).

Source: brochure-v2 · energiacoop.cleantechhub.net/casos/



CURRENT

REIN Hubs

LATAM physical network · PVBLIC Foundation

5+	CLP	CTH
LATAM hubs	Via ClimateLaunchpad	Delivery partner

What we did

- Colombia hub at Universidad Externado (REIN Colombia).
- Cleantech cohorts with public profiles.
- No $ figures in this deck (AGENTS.md).

Source: decks/rein-academic-city-ghana · rein Canva (photo).



PRIOR

P4G / MubOn

Competitive archive · 50/50 co-financing

50/50
P4G co-finance

Grant
UNDP graph · Drive

What we did

- Proposals and MubOn packs in the grant desk archive.
- Electric mobility line as a P4G portfolio example.
- Program traceability only — no award amounts here.

Source: skills/cth-grant · corporate p22 (EV photo).



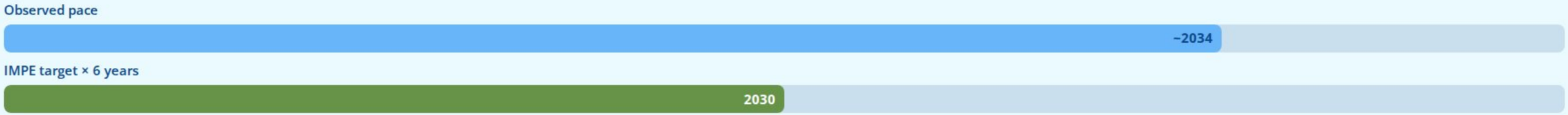
ENERGY COOP

The gap and OPEX

IMPE · ICEE · rural operations

15.4%	9.5%	75.9%	63.1%
IMPE ~8.1M	2030 target	Rural ICEE	IMPE Guajira

Observed pace ~300k/year closes the 2.9M-person gap toward **~2034**; IMPE target ~490k/year points to **2030**. Contraloría, Amazonas, and CREG document unremunerated OPEX on rural microgrids.



CTH: *bankable* social layer and OPEX with committed CAPEX and implementer.

Source: IMPE 2025 · Contraloría · UPME ICEE.



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PROPOSAL

Cooperative + ecosystem

Community O&M · cooperative governance

~1k	100	20	10
Public cross-cut	Scoring	Validation	Co-op wave

IMPE dimension KPIs (does not replace ICEE). Brochure-v2 ecosystem:

Sunco 898 kWp[9] · SEOS +75[10] · Soluna ZNI · Tierra Grata 1,947[11] · TuWatt Coops

Candidate: Corozal, Sucre (IMPE 43.1%) — not a PASO energy project.



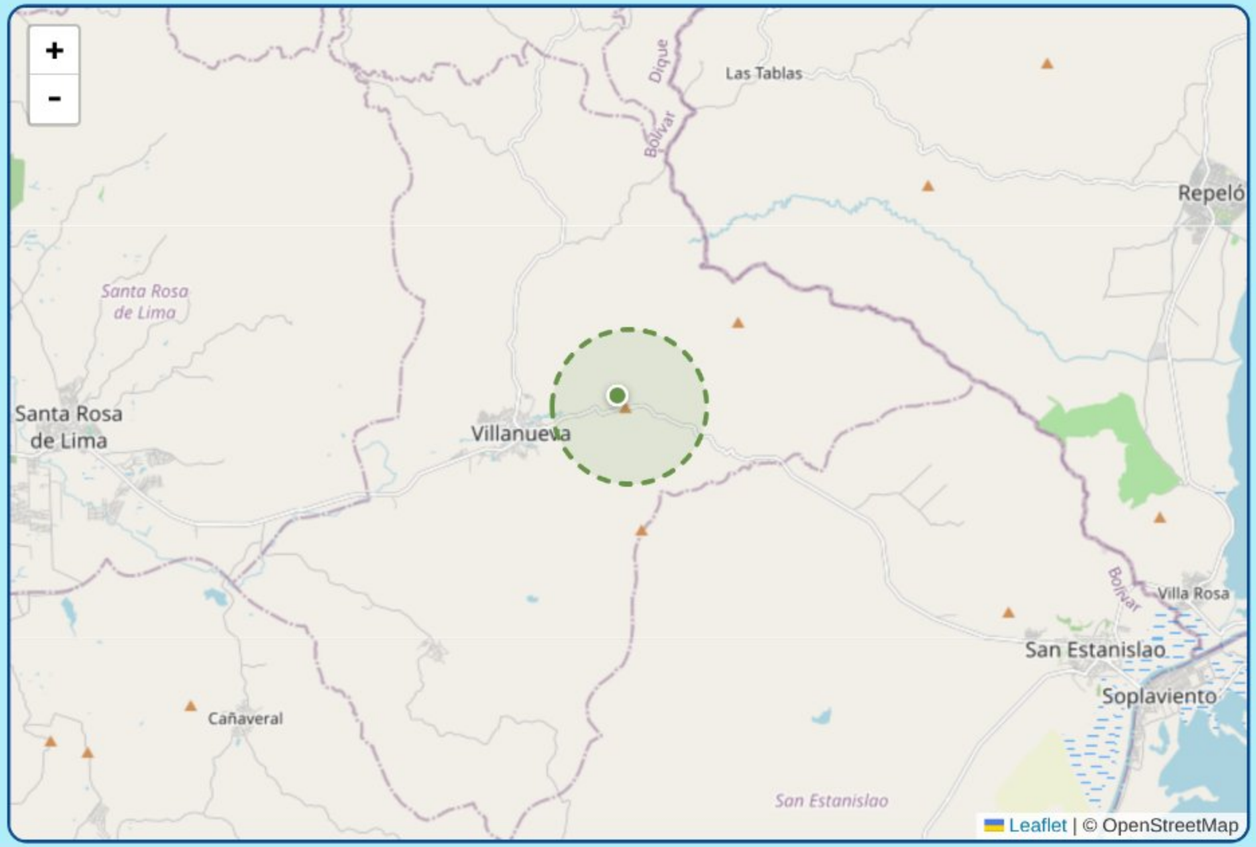
Source: brochure-v2 · illustrative data lab.

Paiva: from committee to cooperative

Community solar energy in vereda Paiva, Bolívar (~450 inhabitants), IE Chiricoco. NOT FENOGE Map location **approximate**.

National energy poverty (IMPE, index on persons): 15.4% · ~8.1 M people (2024)
Informe IMPE 2025 (PDF) · Nota IMPE, nov 2025

Rural electricity coverage 75.92% vs national 93.12% (UPME ICEE 2024)
Boletín ICEE 2024 (UPME) · UPME, noticia 2024



PUBLIC SOURCE

Community solar project with local committee and IE Chiricoco (linked public sources).

Noticia Paiva (2025) · Guía del Gas

CTH DESIGN PROPOSAL

Guided transition: community committee → energy cooperative with ~30% participation in the community share class (CTH proposal, not a contractual figure).

Community committee
Grassroots organization

→

Energy cooperative
Legal personality

Transition shown

- General assembly — supreme social body
- Board of directors — oversight and strategy
- Operator — technical O&M of the asset
- Energy-sector ally — know-how (role per local agreement)
- Community share class — ~30% CTH proposal in capital structure

~450 inhabitants and territory: public sources (links). Map coordinates: approximate (no geocode in source).

La Guajira at scale

Public projects in La Guajira (FENOGE and related processes). Pins without official coordinates: *approximate location*.
Verification: `docs/energy-coop/demo-cth-brand/SOURCES.md`.

IMPE La Guajira 63.1% (2024)
Informe IMPE 2025 (PDF) · PARTIAL — 1ST DEPT. SOURCE

ICEE La Guajira 54.28% (UPME 2024)
Boletín ICEE 2024 · UPME, noticia

Cooking with firewood/charcoal/waste: 56.6% of people in energy poverty (IMPE 2025)
Informe IMPE 2025 (PDF) · Resumen ejecutivo IMPE 2025

4	2,102	750
Projects on map	Verified kWp (sum)	Announced MW (utility-scale)

kWp sum includes only Maicao (1,102) + Cabo/Media Luna (1 MW = 1,000 kWp). Nazareth: kWp not published. 750 MW project is not added to kWp.

4 projects on map



Data Lab — prioritization funnel

How the Energy Coop pipeline narrows candidates before PASO. Public policy figures and documented cross-checks from `docs/energy-coop/`; wave counts (100 → 20 → 10) are EXAMPLE DATA illustrating the flow.

- **Public universe (reference)**
  National target of **500 energy communities 2024–2025**, 40% ZNI (FENOGE). In La Guajira, IPSE reports **148 EC** and **1,409 families** under Miichi Ka'i (Dec 2024).
  Sources: `energy-poverty-kpis-2026-11.md` · FENOGE · IPSE
- **IMPE / ICEE / IPEM cross-check**
  Desk layer: departmental IMPE incidence (e.g. La Guajira 63.1%, Chocó 52.8%), rural vs urban ICEE, household IPEM (different index). Does not replace field visits.
  Cross-check: `energy-poverty-kpis-2026-11.md` · IMPE 2025 PDF
- **Solar resource layer (screening)**
  Google Solar BASE where coverage exists (e.g. Corozal, Uribia); NASA POWER / Global Solar Atlas where not (Quibdó, Bahía Solano). Roof signal, not ground truth.
  `solar-api-coverage-2026-11.md` · `solar-api-integration-plan.md`

EXAMPLE DATA

100
Candidates after public cross-check + preliminary scoring

↓

20
Prioritized after technical and governance validation

↓

10

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PASO — productive community candidates

Territories in dialogue for productive cooperative models. Locations and candidate status only.

CANDIDATE COMMUNITY

Corozal, Sucre

Municipal seat — reference point with Google Solar BASE coverage in map check (9.31778, -75.29583).
`solar-api-coverage-2026-11.md`

Productive communities in dialogue (territorial name only):

CANDIDATE COMMUNITY
Fonseca, La Guajira

CANDIDATE COMMUNITY
Puerto Asís / Guzmán, Putumayo

CANDIDATE COMMUNITY
La Macarena, Meta

CANDIDATE COMMUNITY
Istmina, Chocó

CANDIDATE COMMUNITY
San José del Guaviare, Guaviare

CANDIDATE COMMUNITY
Miranda, Cauca

CANDIDATE COMMUNITY
Arauca, Arauca

Candidate list for PASO conversation; not final selection or financing commitment.

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Your cooperative in 3 clicks

Illustrative CEOF flow (cooperative structure + investors + operator). CTH DESIGN PROPOSAL

1. Choose a community — Select —

2. System size

10 kWp | 25 kWp | 50 kWp | 100 kWp

Create cooperative

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2030 target + local talent

National IMPE scenario. Index on **persons**. Sources: `energy-poverty-kpis-2026-11.md`.

IMPE 2030 target: 15.4% (8.11 M people, 2024) → 9.5% (5.17 M) in 2030 · −2.9 M people
Informe IMPE 2025, p. 33 (PDF) · Nota IMPE, nov 2025

Pace gap (cited arithmetic)

People who must leave energy poverty ≈ **2.9 M** (8.11 M − 5.17 M; IMPE 2025 p. 33).

Observed pace ~300,000 people/year

`2,900,000 ÷ 300,000 ≈ 9.7 years from 2024`

~2034

IMPE 2025 p. 12 (~300 k/year) · IMPE 2025 report (PDF)

Target scenario pace ~490,000 people/year × 6 years

`490,000 × 6 = 2,940,000 ≈ 2.9 M`

2030

IMPE 2025 p. 33 (≥1.0 p.p./year incidence)

Not a CAPEX problem, it is OPEX

- Contraloría: no methodologies to remunerate operation, microgrid administration and SFVI → maintenance and monitoring failures. El Colombiano
- Amazonas (~$30,000 million in royalties): abandoned photovoltaic project; damaged batteries — “critical state” due to sustainability. Seguimiento.co / Procuraduría
- CREG (GEN-Renovables): thousands of rural PV systems without management → “often abandoned”; missing regulation, certified installers and O&M. Circular 005-13 informe final · Concepto CREG 4022/2019 (AOM)

Local talent — documented links

- Escuela Colombiana de Ingeniería Julio Garavito: Luis Carlos Sarmiento Angulo Chair — energy sector axis; academic challenge integrating electricity and gas markets. CLSA sector energético · Primer reto CLSA
- Solucionatón 10X (Connect Bogotá): GeoPrioriza challenge — 3rd place, Escuela students (Oct 2026). Noticia Escuela
- **Caribbean:** scholarships with **Uninorte** (ProExcelencia) Uninorte; **UTB** (Vamos pa’ Lante, campus solar) UTB; **Uniautónoma** — 30

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Live demo

https://energiacoop.cleantechhub.net/demo/

Guided tour · EN/ES · Paiva · La Guajira · data lab · PASO · cooperative · 2030

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